

# PHYS 487: Lecture # 2

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## Lecture Span

- Symmetries; invariance under transformations
- transformations in space
- generators
- translational invariance

## What is a symmetry?

A symmetry exists if applying a transformation to an object will not change anything observable of the object. There exists discrete symmetries (flipping, rotating by  $\pi/2$ ) and continuous symmetries (circle).

## Describing transformations

Suppose

$$A|\psi\rangle = |\psi'\rangle \quad (1)$$

Where  $A$  is some operator that acts on the wave function. If  $A$  acts on a symmetry of the wave function, then its expectation value shouldn't change. That is

$$\langle O \rangle = \langle \psi | O | \psi \rangle = \langle \psi | A^\dagger O A | \psi \rangle \quad (2)$$

If this is true, then we say that  $\langle O \rangle$  is unchanged under the transformation of  $A$ . This formalism is equivalent to a relative transformation, either transforming  $|\psi\rangle$  or  $\hat{O}$ . Eitherway, they are equivalent. We call a transformation of the wavefunction an "active transformation" and the transformation of the observable as a "passive transformation".

## Symmetries + Hamiltonian

A wavefunction is translationally invariant if its Hamiltonian is also translationally invariant.

$$\text{Invariance: } A^\dagger H A = H' = H$$

We first define "unitarity", that is  $A^\dagger A = A A^\dagger = \mathbb{I}$ . That is  $A^\dagger = A^{-1}$ .

$$A^\dagger H A = H$$

$$A A^\dagger H A = A H \text{ (assume } A \text{ is unitary)}$$

$$H A = A H$$

$$[\hat{H}, \hat{A}] = 0$$

This proof is saying what (by default  $A$  is unitary), that if  $\hat{H}$  is invariant under  $A$ , then  $[\hat{H}, \hat{A}] = 0$ .

A unitary is reversible and energy conserving. They also preserve inner product:

$$\langle \psi | A^\dagger A | \psi \rangle = \langle \psi | \psi \rangle \quad (3)$$

## Transformations in space

1. Translation

$$\hat{T}(a)\psi(x) = \psi'(x) = \psi(x - a) \quad (4)$$

2. Rotation

$$R_z(\varphi)\psi(r, \theta, \phi) = \psi(r, \theta, \phi - \varphi) \quad (5)$$

3. Parity: reflection about the origin

$$\hat{\Pi}\psi(r) = \psi(-r) \quad (6)$$

## Translation operator

$$T(a)\psi(x) = \psi(x - a) \quad (7)$$

Maybe momentum? Recall, momentum:  $\hat{p} = -i\hbar\partial_x$ . Let's try to Taylor expand  $T\psi$  by expanding  $\psi(x - a)$ . Recall that a Taylor expansion about  $a$  is

$$f(x) \approx \sum_n \frac{f^{(n)}(a)}{n!} (x - a)^n \quad (8)$$

Then let  $h = x - a$ , then we obtain

$$f(a + h) \approx \sum_n \frac{f^{(n)}(a)}{n!} h^n \quad (9)$$

Note that  $a$  can be  $x$ .

$$\begin{aligned} & \sum_{n=0}^{\infty} \frac{1}{n!} \frac{d^n}{dx^n} \psi(x) ((-a)^n) \\ &= \sum_n \frac{1}{n!} \left( -\frac{ia}{\hbar} \hat{p} \right)^n \psi \\ &= T(a) = \exp \left( -\frac{ia}{\hbar} \hat{p} \right) \end{aligned}$$

We then call  $\hat{p}$  a generator.

## Properties

1. Unitarity:  $T^{-1}(a) = T(-a) = T(a)^\dagger$

if  $\hat{Q}$  is hermitian, then  $u = \exp(iQ)$  is unitary.

2. Action on  $\hat{x}$

$$T^\dagger(a) \hat{x} T(a) = \hat{x} + a = \hat{x}' \quad (10)$$

$$\hat{x}' \psi(x) = T^\dagger(a) \hat{x} T(a) \psi(x) \quad (11)$$

$$= T(-a) x \psi(x - a) \quad (12)$$

$$= (x + a) \psi(x) \text{ (assuming that } \psi \text{ is invariant under } T) \quad (13)$$

In general,  $T^\dagger Q(x, p) T = Q(x + a, p)$ . Note, in more dimensions, stepping  $\Delta x_i$  in any dimension wouldn't change the observable of the object. If  $[A, B] = 0$ , then  $\exp(A + B) = \exp(A) \exp(B)$ . But if  $[A, B] \neq 0$ , then there is a geometric phase acquired when stepping  $\Delta x_i$ .

## Translational Symmetry

1. Continuous: free particle
2. Periodic (and infinite)

Consider a infinitely periodic potential with a period. Then consider

$$H' = T^\dagger(a)HT(a) = T^\dagger(a) \left( \frac{p^2}{2m} + V(x) \right) T(a) \quad (14)$$

$$= \frac{p^2}{2m} + T^\dagger(a)V(x)T(a) = \frac{p^2}{2m} + V(x+a) \quad (15)$$

## Bloch's Theorem

Know:  $[\hat{H}, \hat{T}] = 0$ . And assume,

$$H\psi(x) = E\psi(x) \quad (16)$$

Then because  $H$  and  $T$  commute, they share a simultaneous set of eigenvalues. Therefore, assume

$$T(a)\psi(x) = \lambda\psi(x) \quad \lambda \in \mathbb{C} \quad (17)$$

But since  $T$  preserves the inner product,

$$\langle \psi | T^\dagger T | \psi \rangle = \langle \psi | \psi \rangle \quad (18)$$

$$\lambda^2 = \langle \psi | \psi \rangle \implies \lambda = \exp(i\phi) \quad \phi \in \mathbb{R} \quad (19)$$

Let  $\phi = qa$ , then

$$\psi(x-a) = e^{-iqa}\psi(x) \quad (20)$$

We can call  $\hbar q$  the "crystal momentum". Then let  $\psi(x) = e^{iqx}u(x)$

$$e^{-iqa}e^{iqx}u(x) = e^{iq(x-a)}u(x) \quad (21)$$

Then shift by  $a$

$$\psi(x) = e^{iqx}u(x+a) \quad (22)$$

Thus, we obtain

$$e^{iqx}u(x) = e^{iqx}u(x+a) \quad (23)$$

This means that  $u$  is periodic in  $a$ .